

Generative Artificial Intelligence (AI)

Acknowledgement

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I acknowledge the use of [1] ChatGPT (<https://chat.openai.com/>) to [2] support preparation for my interview, provide guidance on academic writing and referencing, assist with version control using GitLab and RStudio, and review and give feedback on my interim report. The prompts (generated between November and January 2026) used include:

[3]

1. “Based on my report and interim report brief what are some questions that can be asked during my interview.”
2. “What is the best way to do a literature review?”
3. “I have a project in GitLab and I have an Rmd file in RStudio, I want to add it to my GitLab, how?”
4. “(Hu and Sokolova 2020) assessed 6 machine learning models... is it like this how I am supposed to reference using Harvard referencing or like Hu et al?”
5. “Does appendix go in the table of contents?”
6. “How is my report, rate it.”

The prompt outputs are included in Appendix A at the end of the assessment [4].

The output from the generative artificial intelligence was [5] used as a guide and a double check only all written content, analysis, and final wording in this assessment are my own.

Appendix A:

Prompt 1:

And honestly... your project is **very solid and well thought-out**. The panel will absolutely go deeper than generic questions with you because your design (data reduction + clustering + two-stage hurdle model) is more advanced than most undergrad projects.

Below are **high-probability, tailored Q&A questions** based *specifically* on your report content, rubric, and what examiners love to probe.

◆ 1. “Elevator Pitch” (They almost always start here)

Q1. Can you summarise your project in under one minute?

Q2. What gap in the literature does your project address?

Q3. Why did you choose this dataset and problem?

Q4. What makes your approach novel compared to existing studies?

Expect them to push you to say:

“Most studies do binary prediction. I use a 3-class + two-stage hurdle model + clustering + mixed-type data reduction.”

◆ 2. Research Questions & Hypothesis

Q5. Walk us through your three research questions and how they connect.

Q6. Why is data reduction your *first* research question?

Q7. How exactly does clustering help prediction in your design?

Q8. Why is a two-stage hurdle model more appropriate than a single 3-class classifier?

Q9. What would it mean if RQ1 succeeds but RQ2 fails?

Q10. What would falsify your hypothesis?

◆ 3. Dataset & Ethics (Very likely)

Q11. Where does your dataset come from and how was it originally collected?

Q12. Why is this dataset suitable for readmission prediction?

Q13. What ethical risks exist in this project?

Q14. Why did you still need ethical approval even though the dataset is public?

Q15. How does GDPR relate to your work?

Q16. What are the dangers of using EHR data for ML?

◆ 4. Data Cleaning Decisions (They WILL test these)

Q17. Why did you drop *weight*, *medical_specialty*, and *payer_code* instead of imputing them?

Q18. Why was it acceptable to drop rows with missing race/diagnosis?

Q19. Why didn't you transform skewed numerical variables?

- Q20. Explain your decision to remove duplicate patients.
- Q21. What is data leakage and how did you avoid it?
- Q22. Why did you map 700 diagnosis codes into 9 categories?
- Q23. Which dropped feature might later turn out to be important?

◆ **5. Exploratory Data Analysis (They love figures)**

- Q24. What is the most important insight from your EDA?
- Q25. Why do early-readmitted patients have longer hospital stays?
- Q26. Why do they have more medications and lab procedures?
- Q27. What does the class imbalance imply for modelling?
- Q28. Why are low correlations not necessarily “good news”?
- Q29. Which variable do you expect to be the strongest predictor and why?

◆ **6. Literature Review (They’ll test depth, not memory)**

- Q30. Which paper influenced your project design the most?
- Q31. Why do many papers get only ~65% accuracy?
- Q32. Why do some deep learning papers show unrealistically high accuracy?
- Q33. Why did you *not* choose an LSTM?
- Q34. What limitations of SMOTE concern you?
- Q35. How does your approach improve on Pham et al. (2019)?
- Q36. Why do different studies contradict each other on DL vs ML?

◆ **7. Data Reduction & Clustering (Your “unique” angle)**

- Q37. Why did you choose FAMD and MCA specifically?
- Q38. Why not PCA alone?
- Q39. How will you decide how many components to keep?
- Q40. How will you evaluate cluster quality?
- Q41. Why k-means AND k-prototypes AND LCA?
- Q42. What if clusters are not meaningful?
- Q43. How will you use clusters in prediction?

◆ **8. Two-Stage Hurdle Model (THIS is your core defence)**

- Q44. Explain the two-stage hurdle model in simple terms.
- Q45. Why is a single 3-class classifier mathematically flawed here?
- Q46. How will you train stage 2?
- Q47. What happens if stage 1 makes an error?
- Q48. How will you evaluate the full system fairly?
- Q49. Why not hierarchical classification instead?
- Q50. What metrics matter most for your problem and why?

◆ **9. Practical ML Decisions**

Q51. Why these models: RF, XGBoost, LightGBM, KNN, Logistic?

Q52. Why not SVM or deep neural nets?

Q53. How will you tune hyperparameters?

Q54. How will you handle class imbalance?

Q55. How will you avoid overfitting?

Q56. What baseline will you compare against?

Q57. What result would you consider “good enough”?

◆ 10. Risks, Limitations & Future Work

Q58. What is the biggest risk in your project right now?

Q59. What will you do if clustering doesn’t help prediction?

Q60. What if the hurdle model underperforms a simple model?

Q61. What part of your pipeline is most fragile?

Q62. How would you extend this into a Master’s project?

Q63. What would you change if you had real hospital data?

◆ 11. Version Control & Engagement (Easy Marks)

Q64. How often did you meet your supervisor and what changed because of feedback?

Q65. Show us an example of incremental development in your repo.

Q66. What was your biggest technical challenge so far?

Q67. What’s one mistake you’re glad you made?

What I Recommend Next (To Lock in 5/5)

If you want, I can:

 Write **perfect model answers** for:

- Elevator pitch
- Two-stage hurdle model
- Data cleaning justification
- Ethics defence
- Why your project is novel
- Biggest limitation + fallback plan

Tell me which **5–7 questions** you’re most nervous about and I’ll script *panel-grade* answers tailored exactly to *your* report and marking rubric.

We can basically rehearse your Q&A until it’s bulletproof. 

Prompt 2:

A **strong literature review** is not about summarising papers one by one — it's about **telling a structured story of what is known, what is debated, and what is missing**. Here's the **best-practice approach**, step by step (the way lecturers and journals expect it).

1 Be clear on your purpose (before reading)

Ask yourself:

- **Why** am I doing this review?
 - Background for a project?
 - Justifying a research gap?
 - Comparing methods?
- **What is the scope?**
 - Time range (e.g. last 5–10 years)
 - Domain (e.g. medical imaging, synthetic data, ML)

👉 Write **1–2 guiding research questions**.

Example:

How effective is synthetic data generation for improving medical image classification performance?

This stops your review from becoming unfocused.

2 Search strategically (not randomly)

Use **academic sources only**:

- Google Scholar
- IEEE Xplore
- ScienceDirect
- SpringerLink
- PubMed (medical/health)
- arXiv (for ML/AI – still cite carefully)

Search tips:

- Use **keyword combinations**
 - "synthetic data" AND "medical imaging"
 - "GAN" AND "lung cancer" AND "CT"
- Use **citation chaining**
 - Read a strong paper → check *its references*
 - Check *who cited it* (very powerful)

👉 Save papers immediately (Zotero / Mendeley / EndNote).

3 Read in layers (this saves time)

Don't fully read everything at first.

First pass (5–10 mins per paper):

- Abstract
- Introduction
- Conclusion

Ask:

- Is this relevant?
- What method/problem/result does it address?

Second pass (important papers only):

- Methods
- Key results
- Limitations

👉 Create short notes:

- **Problem**
- **Method**
- **Dataset**
- **Key findings**
- **Limitations**

Group papers by themes (CRUCIAL)

This is where most students go wrong.

 Wrong:

Paper 1 → Paper 2 → Paper 3 summary

 Correct:

Organise by **themes**, not papers.

Examples of themes:

- Types of methods (GANs, VAEs, diffusion models)
- Applications (classification, segmentation, augmentation)
- Evaluation strategies
- Limitations (bias, realism, privacy)
- Medical domains (lung cancer, MRI, CT)

 Each paragraph = **one theme**, supported by multiple papers.

Be analytical, not descriptive

For every theme, answer:

- What do studies **agree on**?
- Where do they **disagree**?
- What are the **limitations**?
- What is **missing**?

Example sentence:

While several studies report improved classification accuracy using GAN-generated data, many rely on small datasets and lack external validation, limiting generalisability.

This shows **critical thinking** (very important for high grades).

Identify the research gap

Your review should naturally lead to:

“This is what is missing → this is what my work addresses”

Gaps can be:

- Limited datasets
- Poor evaluation metrics
- Lack of comparison between models
- Focus on one class only (e.g. malignant but not benign)
- No clinical validation

👉 Your final paragraph should clearly state the gap.

7 Structure of a high-quality literature review

A common structure:

1. Introduction

- Context
- Importance of the topic
- Aim of the review

2. Thematic sections

- Methods
- Applications
- Evaluation
- Challenges

3. Limitations & gaps

- Weaknesses in existing work

4. Summary & link to your study

- How your work fits in
-

8 Write while reviewing (not after)

Don't wait until all papers are read.

- Write rough paragraphs early

- Refine as you read more
- Merge ideas across papers

👉 This avoids last-minute stress and improves coherence.

9 Referencing & academic tone

- Use consistent citation style (APA / Harvard)
 - Avoid:
 - “I think...”
 - “This paper is very good...”
 - Use:
 - “Studies suggest...”
 - “Prior work indicates...”
 - “However, existing approaches...”
-

10 Final quality check

Before submitting, ask:

- Does this review **tell a story**?
 - Are papers **connected**, not isolated?
 - Is there a **clear gap**?
 - Does it justify **my project**?
-

Since you're working on synthetic data & medical imaging, I can:

- Help you **define themes**
- Create a **literature review outline**
- Help you **rewrite a section critically**
- Check if your review is **first-class level**

Just tell me what stage you're at 💙

Prompt 3:

Got it! You want to **add your .Rmd file from RStudio to a GitLab project**. I'll walk you through the steps carefully. There are **two main ways**: using **RStudio's Git integration** or using **Git commands in the terminal**.

Option 1: Using RStudio Git Integration (easiest)

1. **Make sure your RStudio project is linked to Git**
 - Open your RStudio project.
 - Go to Tools → Project Options → Git/SVN.
 - Make sure **Git** is enabled.
 - If your project is not a Git repo yet, you can initialize Git: Tools → Version Control → Project Setup → Git.
 2. **Clone your GitLab repo into RStudio** (if you haven't already)
 - Copy the HTTPS/SSH URL of your GitLab repo.
 - In RStudio: File → New Project → Version Control → Git
 - Paste your GitLab URL and choose a local folder.
 - This will clone the repo to your machine and link it with RStudio.
 3. **Add your .Rmd file**
 - Move your .Rmd file into the project folder (or create it inside RStudio in the project folder).
 4. **Commit the changes**
 - Go to the **Git tab** in RStudio (usually top-right).
 - You should see your .Rmd file listed as untracked.
 - Check the box next to the file → Click **Commit**.
 - Add a commit message (e.g., "Add initial R Markdown file") → Click **Commit**.
 5. **Push to GitLab**
 - After committing, click **Push** in the Git tab.
 - Your .Rmd file is now in GitLab.
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Option 2: Using Git commands in the terminal

If you prefer the terminal:

1. **Open terminal in your project folder** (RStudio has a terminal tab, or use your system terminal).
2. **Initialize Git** (if not already):
3. `git init`
4. **Link your GitLab repo** (replace the URL with your repo's URL):
5. `git remote add origin https://gitlab.com/username/projectname.git`
6. **Add your .Rmd file:**
7. `git add your_file.Rmd`
8. **Commit the changes:**
9. `git commit -m "Add initial R Markdown file"`
10. **Push to GitLab:**
11. `git push -u origin main`

(replace main with your default branch if it's different)

Tips:

- Make sure your .Rmd file is inside the folder that is connected to Git.
- If Git is not installed or RStudio doesn't detect it, you need to [install Git](#) and configure it:
- `git config --global user.name "Your Name"`
- `git config --global user.email "you@example.com"`

If you want, I can give you a **super quick RStudio step-by-step with screenshots in mind** that makes it literally 3 clicks to push your .Rmd to GitLab.

Do you want me to do that?

Prompt 4:

Good question! It depends on **how many authors there are** in the original paper.
Harvard-style rules are:

1 Two authors

- **In-text citation:** list both names every time.
- Example: (Hu and Sokolova 2020) ✓ correct.

So your sentence is correct **if the paper has exactly 2 authors**.

2 Three or more authors

- **In-text citation:** use the first author + *et al.*
 - Example: (Hu et al. 2020)
 - Reference list: list **all authors** (usually up to 6; if more, sometimes “first 3 then et al.” depending on your style guide)
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✓ Your sentence

If Hu & Sokolova **are the only authors**, this is correct:

“Hu and Sokolova (2020) assessed 6 machine learning models including Random Forest, decision trees, SVM and Logistic Regression for a multi-class prediction of readmission, which is a three-category variable.”

If there are **3+ authors**, change it to:

“Hu et al. (2020) assessed ...”

If you want, I can **check all your references you’ve collected** and tell you **exactly which should be “et al.” and which should list all authors**, so your Harvard in-text citations are 100% correct.

Do you want me to do that?

Prompt 5:

Yes — **appendices should absolutely go in your Table of Contents (ToC)**, especially in academic reports like your interim report.

Here’s why and how:

 Why appendices should be in the ToC

1. **Professionalism:**

Including appendices shows that your report is **complete, structured, and organized**.

2. **Rubric alignment:**

Your rubric emphasizes **document structure and writing (5 marks)** and mentions ethical approval, AI prompts, code, and journals — all of which go in appendices. A ToC makes these **easy for assessors to find**, which helps your score.

3. **Navigation:**

Appendices are usually **after the references**. Listing them in the ToC makes it easy for readers to jump straight to them.

 How to include appendices in your ToC

- Label your appendices clearly: Appendix A, Appendix B, etc.
- Include a **short descriptive title** for each appendix.
- Example in ToC:

References 20

Appendices 22

 Appendix A: Ethical Approval Documentation 22

 Appendix B: Generative AI Usage and Prompts 23

 Appendix C: Code Repository Link 24

 Appendix D: Project Journal / Meeting Minutes 25

- The page numbers should match your document.
-

 Tip

Even though appendices are “extra material,” **always include them in the ToC** in academic reports. It’s expected at the BSc level and shows attention to detail.

If you want, I can **draft a complete Table of Contents for your interim report** including all main sections **and appendices**, fully rubric-aligned and ready to paste into Word or PDF.

Do you want me to do that?

Prompt 6:

Thanks for sharing your full report, Israa! I've gone through it carefully. Overall, it's a **strong, comprehensive, and well-structured interim report**—you clearly know your dataset, methods, and research goals. Here's a detailed evaluation with constructive feedback:

Strengths:

1. Structure & Organization:

- Clear sections: Introduction, Literature Review, Data Exploration, Proposed Future Analysis, References, and Appendix.
- Table of Contents is precise, and headings are logical.
- Research questions and hypotheses are clearly stated.

2. Content Depth:

- Introduction shows strong understanding of the problem (diabetes, readmissions, costs).
- Literature Review is thorough and critically analyzes existing studies.
- Data exploration and cleaning process is detailed and methodical.
- Ethical approval form is fully included in Appendix.

3. Technical Knowledge:

- You demonstrate excellent knowledge of machine learning techniques (clustering, two-stage hurdle model, Random Forest, XGBoost, LSTM, etc.).
- Data cleaning decisions are well justified (handling missing values, outliers, duplicates, low-variance columns).
- EDA visualizations and summary statistics are well thought out.

4. Professional & Academic Tone:

- Generally consistent formal tone.

- References are properly cited using Harvard style (mostly).

5. Forward Planning:

- Proposed Future Analysis section shows a realistic timeline.
 - Clear link between data preparation, feature engineering, clustering, and predictive modelling.
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Areas for Improvement:

1. Grammar & Style:

- Some sentences are very long and run-on. For example:

“Accidental readmission increase both patient financial strain as well as inefficient use and misallocation of medical resources that is why and in our current era due to the rapid growth that the AI field has experienced...”

Consider breaking long sentences into 2–3 shorter sentences for clarity.

- Minor typos:
 - “Americas” → “Americans”
 - “inpatient readmissions is usually based the length” → “inpatient readmissions are usually based on the length”
 - “ius why” → “is why”

2. Citations:

- Consistency: Some citations are (Hu and Sokolova 2020) while others are (Sah Kanu and Khanal 2023) or missing parentheses entirely.
- Make sure **all in-text citations match your reference list** and follow Harvard style consistently.
- Some citations can appear **at the end of paragraphs**, not just at the start, especially when summarizing multiple points.

3. Figures & Tables:

- You list figures (Fig.1, Fig.2 ...) but don't include the actual visuals in the text you shared. Ensure figures are **properly inserted with captions** in the final Word/PDF.
- Table 1 is clear, but consider **consistent formatting** for numeric values (e.g., 2 decimal places).

- All figures and tables should be **referenced in text** (e.g., “As shown in Fig. 3, the readmission distribution by gender shows...”).

4. **Clarity & Redundancy:** ⚠️

- Some parts repeat ideas (like importance of clustering and data reduction). Tighten by merging sentences to reduce redundancy.
- E.g., “Clustering analysis to discover subgroups ... as well as data reduction analysis ...” could be shortened to “Clustering analysis and data reduction will be used to identify patient subgroups and key features.”

5. **References Formatting:** ⚠️

- Some entries are missing journal/volume/page info (e.g., Hu and Sokolova 2020).
- Check spacing and punctuation: some references have extra commas or missing periods.

6. **Minor Formatting:** ⚠️

- Introduction and Literature Review are dense. Use **subheadings** for clarity (e.g., “Machine Learning Approaches,” “Deep Learning Approaches”).
- Ensure consistent font, line spacing, and alignment throughout.

Overall Rating:

I would rate your interim report as **Very Good / 85–90%** for an undergraduate research project.

- **Content:** Excellent
- **Technical understanding:** Excellent
- **Structure & organization:** Good–Very Good
- **Writing clarity & grammar:** Moderate (needs minor polishing)
- **Formatting & citations:** Moderate (needs consistency)

💡 **Tip for improvement before submission:**

1. Go through the report sentence by sentence and break very long sentences.
2. Check all citations for consistency.
3. Insert figures properly with captions and references.

4. Proofread for small typos and grammar issues.

If you want, I can **make a polished, “submission-ready” version** of your Introduction and Literature Review with corrected grammar, citations, and sentence structure — it would read **much cleaner and professional** without changing your content.

Do you want me to do that?